

MetaVista Advanced Research Report

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Executive Summary

This report was generated through a modular multi-agent workflow. The system assessed the prompt, activated relevant specialist agents, collected outputs, reviewed risks, and synthesized a detailed research-style answer.

Topic: Not provided

Acting Lens: university research advisor

Research Depth: Medium Research

User Prompt

Act as a university research advisor. Provide a detailed research proposal outline for the topic: "The impact of AI-generated content on student critical thinking skills in higher education."

Include the following sections:

1. Introduction: Define the problem statement and research significance.
2. Literature Review Themes: Suggest three current themes or debates in academia regarding AI in education.
3. Methodology: Propose a mixed-methods research design (qualitative and quantitative) suitable for this study.
4. Data Collection: Detail how to ethically gather data from students and professors.
5. Potential Limitations: Identify two major hurdles or biases this research might face.

Keep the tone academic, structured, and actionable.

Activated Workflow

1. Assess the prompt.
2. Select relevant agents.
3. Run specialist agents.
4. Review assumptions and evidence gaps.
5. Synthesize final report.
6. Generate downloadable PDF if enabled.

Specialist Findings

EX Information Extraction Agent

Information Extraction Agent

Topic: Not provided
Acting Lens: university research advisor
Depth: Medium Research

Extracted Task

- Main request: Act as a university research advisor. Provide a detailed research proposal outline for the topic: "The impact of AI-generated content on student critical thinking skills in higher education."

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- Required output: advanced research-style answer
- Needs: structure, evidence awareness, recommendations, and final report readiness.

Decomposition

1. Understand the task.
2. Identify facts, assumptions, constraints, and expected outputs.
3. Separate evidence-backed claims from reasoning.
4. Prepare clean sections for synthesis.

TB Tables and Comparison Agent

Tables and Comparison Agent

Suggested Tables

| Table | Purpose |

|---|---|

| Agent activation matrix | shows which agents were used and why |

| Evidence quality matrix | separates strong and weak claims |

| Risk table | exposes uncertainty and mitigations |

| Recommendation table | converts research into decisions |

SK Skeptical Review Agent

Skeptical Review Agent

Risks

- Multi-agent output can still hallucinate.
- PDF formatting can make weak research look authoritative.
- Deep research may become slow or expensive.
- Agents may duplicate each other unless prompts are distinct.

Safeguards

1. Router chooses only relevant agents.
2. Reference agent flags unsupported claims.
3. Synthesizer preserves caveats.
4. Final report separates facts from recommendations.

MS Master Synthesis Agent

Master Synthesis Agent

The final output should be a polished advanced report with:

- executive summary
- task framing
- activated agents
- specialist findings
- evidence gaps
- tables/visual suggestions
- final recommendation

Tone: university research advisor

Depth: Medium Research

RF Reference and Evidence Agent

Reference and Evidence Agent

Evidence Strategy

- Prefer official docs, primary sources, research papers, and credible technical references.
- Mark unsupported claims clearly.
- Use citations for current facts, laws, prices, model specs, and benchmarks.
- Separate verified evidence from assumptions.

Citation Plan

| Claim Type | Source Type | Rule |

|---|---|---|

| Definition | official docs/textbook | cite once |

| Current fact | recent web source | cite every claim |

| Technical claim | docs/paper | cite version/date |

| Recommendation | source + reasoning | state assumptions |

Final Recommendation

Keep the system registry-based. Each new agent should be added inside `agents/registry.py` with an ID, name, keywords, system prompt, and runner. The orchestrator should only select and call registered agents so future expansion remains simple.

Caveats

This implementation gives you the architecture and UI flow. For production-grade references, connect the Reference Agent to live web search, PDFs, papers, or internal document stores.